

Differential proteomic analysis on the effects of 2-methoxy-1,4-naphthoquinone towards MDA-MB-231 cell line

Kitson Liew^a, Phelim Voon Chen Yong^a, Visweswaran Navaratnam^a, Yang Mooi Lim^b, Anthony Siong Hock Ho^{a,*}

^a School of Biosciences, Taylor's University, No.1 Jalan Taylor's, 47500 Subang Jaya, Selangor Darul Ehsan, Malaysia

^b Department of Pre-Clinical Sciences, Faculty of Medicine and Health Sciences, Universiti Tunku Abdul Rahman, Lot PT21144, Jalan Sungai Long, Bandar Sungai Long, 43000 Kajang, Selangor Darul Ehsan, Malaysia

ARTICLE INFO

Article history:

Received 23 October 2014

Revised 31 January 2015

Accepted 5 March 2015

Keywords:

2-Methoxy-1,4-naphthoquinone

S100A4

RPSA

ERK1/2

NF- κ B

E-cadherin

ABSTRACT

Background: We have previously reported the anti-metastatic effects of 2-methoxy-1,4-naphthoquinone (MNQ) against MDA-MB-231 cell line.

Purpose: To investigate the molecular mechanism underlying the anti-metastatic effects of MNQ towards MDA-MB-231 cell line via the comparative proteomic approach.

Study design/methods: Differentially expressed proteins in MNQ-treated MDA-MB-231 cells were identified by using two-dimensional gel electrophoresis coupled with tandem mass spectrometry. Proteins and signalling pathways associated with the identified MNQ-altered proteins were studied by using Western blotting.

Results: Significant modulation of MDA-MB-231 cell proteome was observed upon treatment with MNQ in which the expressions of 19 proteins were found to be downregulated whereas another eight were upregulated (>1.5 fold, $p < 0.05$). The altered proteins were mainly related to cytoskeletal functions and regulations, mRNA processing, protein modifications and oxidative stress response. Notably, two of the downregulated proteins, protein S100-A4 (S100A4) and laminin-binding protein (RPSA) are known to play key roles in driving metastasis and were verified using Western blotting. Further investigation using Western blotting also revealed that MNQ decreased the activations of pro-metastatic ERK1/2 and NF- κ B signalling pathways. Moreover, MNQ was shown to stimulate the expression of the metastatic suppressor, E-cadherin.

Conclusion: This study reports a proposed mechanism by which MNQ exerts its anti-metastatic effects against MDA-MB-231 cell line. The findings from this study offer new insights on the potential of MNQ to be developed as a novel anti-metastatic agent.

© 2015 Elsevier GmbH. All rights reserved.

Introduction

Metastasis accounts for more than 90% of cancer-related deaths globally (Sethi and Kang 2011). Metastatic disease is characterised

by the acquisition of invasive behaviour by cancer cells in order to migrate to distant organ sites via the vasculature and to subsequently develop into secondary tumours (Sethi and Kang 2011). Thus far, the lack of progress in preventing metastasis can be attributed to the scarcity of anticancer drugs that can successfully combat metastatic progression (Weber 2013). Plant natural products have and will continue to be an important source of novel and effective anticancer drugs. Numerous phytochemicals have shown promising anti-metastatic activities in recent years as they are increasingly reported to modulate proteins and intracellular signalling pathways that govern metastasis (Shu et al. 2010). 1,4-Naphthoquinone derivatives such as plumbagin and shikonin are one such group of phytochemicals that have strong anticancer and anti-metastatic activities (Lu et al. 2013).

2-Methoxy-1,4-naphthoquinone (MNQ) (Fig. 1) is a naturally occurring phytochemical in the plant *Impatiens balsamina* L. Previous studies on the anticancer activities of this 1,4-naphthoquinone derivative are mostly focused on its cytotoxic properties via the

Abbreviations: 2DGE, two-dimensional gel electrophoresis; CHAPS, 3-[(3-cholamidopropyl)dimethylammonio]-1-propanesulfonate; AP-1, activator protein-1; DTT, dithiothreitol; DMSO, dimethyl sulphoxide; EMT, epithelial-to-mesenchymal transition; ERK1/2, extracellular signal-regulated kinase 1/2; FAK, focal adhesion kinase; FBS, fetal bovine serum; IPG, immobilised pH gradient; MALDI-TOF/TOF-MS, matrix-assisted laser desorption ionisation tandem time-of-flight mass spectrometry; MAPK, mitogen-activated protein kinase; MMP, matrix metalloproteinase; MNQ, 2-methoxy-1,4-naphthoquinone; NF- κ B, nuclear factor- κ B; NMIIA, nonmuscle myosin-IIA; ROS, reactive oxygen species; RPSA, laminin-binding protein; S100A4, protein S100-A4; uPa, urokinase plasminogen activator.

* Corresponding author. Tel.: +60 3 56295440; fax: +60 3 56295455.

E-mail addresses: kitson_liew@hotmail.com (K. Liew), phelim.yong@taylors.edu.my (P.V.C. Yong), nava.vis@taylors.edu.my (V. Navaratnam), ymlim@utar.edu.my (Y.M. Lim), anthony.ho@taylors.edu.my (A.S.H. Ho).

<http://dx.doi.org/10.1016/j.phymed.2015.03.007>

0944-7113/© 2015 Elsevier GmbH. All rights reserved.

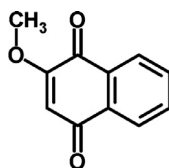


Fig. 1. 2-Methoxy-1,4-naphthoquinone (MNQ).

induction of apoptosis and necrosis (Ding et al. 2008; Tan 2011, Wang and Lin 2012). Nonetheless, we have recently demonstrated the potential of MNQ as a promising anti-metastatic lead compound (Liew et al. 2014). The suppression of invasion and migration of the highly invasive MDA-MB-231 cancer cell line was observed when MNQ was used at minimally cytotoxic concentrations ($\leq 7.5 \mu\text{M}$). Moreover, we have also shown that MNQ could exert its anti-metastatic effects at least via the downregulation of a pivotal metastatic mediator, matrix metalloproteinase-9 (MMP-9) (Liew et al. 2014). The detailed molecular mechanism underlying all these effects however has not been investigated.

Comparative proteomic analysis represents a powerful approach that is frequently employed to elucidate the mode of action of anticancer drug candidates (Kraljevic et al. 2006). Proteins are principal targets of drug discovery since these biomolecules functionally govern cellular processes and ultimately dictate the biological phenotypes. Up- or downregulation of proteins responsible for the pathogenesis of cancer upon a drug treatment are the fundamental requirement in identifying novel and effective targets (Kraljevic et al. 2006). Furthermore, the altered proteins offer clues on the signalling proteins and pathways affected by a drug (Guo et al. 2013).

In this study, we aimed to investigate the modulatory effects of MNQ towards MDA-MB-231 cell proteome by using a comparative proteomic approach. Two-dimensional gel electrophoresis (2DGE) combined with matrix-assisted laser desorption ionisation tandem time-of-flight mass spectrometry (MALDI-TOF/TOF-MS) were used to identify critical target proteins of MNQ. In addition, we examined the effects of MNQ on the intracellular signalling pathways related to these target proteins which could ultimately shed light on the mode of action of MNQ in exerting its anti-metastatic potential.

Materials and methods

Materials

2-Methoxy-1,4-naphthoquinone (MNQ) (98% pure) and dimethyl sulphoxide (DMSO) were purchased from Sigma–Aldrich (St. Louis, MO, USA). The master stock solution of MNQ was prepared at the concentration of 40 mM in DMSO. The subsequent working stocks were prepared at 7.5 mM by diluting the master stock with appropriate amount of DMSO. The tubes used to store these stocks were wrapped in aluminium foil and kept at -20°C . The chemicals and reagents for the whole cell lysate extraction for 2DGE experiment were obtained from Bio Basic Inc (Canada) unless stated otherwise.

Rabbit monoclonal antibodies against protein S100-A4 (S100A4) (ab124805) and laminin-binding protein (RPSA) (ab133645) were purchased from Abcam (Cambridge, UK). Mouse monoclonal antibodies against β -actin (#3700), total ERK1/2 (#9107) and phospho-ERK1/2 (Thr202/Tyr204) (#9106) as well as rabbit monoclonal antibodies against E-cadherin (#3195), total NF- κ B p65 (#4764) and phospho-NF- κ B p65 (Ser536) (#3033) were purchased from Cell Signalling Technology (Beverly, MA, USA). IRDye 680RD goat anti-mouse and goat anti-rabbit IgG secondary antibodies were purchased from LI-COR Biosciences (Nebraska, USA).

Cell culture and treatment

Human breast adenocarcinoma cell line MDA-MB-231 was obtained from American Type Culture Collection (ATCC), USA. Cells were maintained in RPMI 1640 medium (Sigma, USA) supplemented with 10% FBS (JRS Scientific, CA, USA) at 37°C , 5% CO_2 . Cells were detached through trypsinisation and subcultured for every two days. For cell treatment, MDA-MB-231 cells were seeded at the density of 3×10^5 cells per each 60 mm culture dish for 48 h. The cells were then incubated in 5% FBS medium supplemented with either DMSO (control) or $7.5 \mu\text{M}$ of MNQ for 24 h as described previously (Liew et al. 2014). Next, the cells were harvested via trypsinisation and the cell pellets were washed with Tris buffered sucrose (10 mM Tris-HCl, 250 mM sucrose pH 7.0) three times before they were stored at -80°C .

Sample preparation

Cell pellets were lysed in 7 M urea, 2 M thiourea, 4% w/v CHAPS, 65 mM DTT, 2% v/v of Biolyte 3/10 Ampholyte 40% (Bio-Rad, Hercules, CA, USA), 1% v/v protease inhibitor cocktail (GE Healthcare, USA) and 1% v/v nuclease mix (GE Healthcare, USA) at room temperature for 1 h with occasional vortexing. The mixture was then centrifuged at 13,200 rpm for 20 min at 10°C and the whole cell lysate supernatant was collected and stored at -80°C prior to isoelectric focusing (IEF).

2DGE

The protein concentration of the whole cell lysate was determined using Bradford assay (Sigma, USA). Each ReadyStripTM IPG strip (pH 3–10 nonlinear, 17 cm) (Bio-Rad, USA) was passively rehydrated overnight at room temperature with 120 μg of protein sample, which was earlier obtained through appropriate dilution with rehydration buffer (7 M urea, 2 M thiourea, 4% w/v CHAPS, 0.5% v/v of Biolyte 3/10 Ampholyte 40%, 20 mM DTT and 0.002% w/v bromophenol blue). For the first dimension separation, IEF was carried out on the PROTEAN IEF Cell (Bio-Rad, USA) at 20°C as follows: 250 V at linear mode for 20 min, 10,000 V at linear mode for 2.5 h and 10,000 V at rapid mode at 40,000 V h. The minimum current allowed for each strip throughout the focusing was 50 μA .

The focused strips were reduced for 15 min in equilibration buffer (6 M urea, 50 mM Tris-HCl pH 8.8, 30% v/v glycerol, 2% w/v SDS and 0.002% w/v bromophenol blue) containing 2% w/v DTT and alkylated for another 15 min in the same equilibration buffer containing 2.5% w/v iodoacetamide (GE Healthcare, USA) instead of DTT. These equilibration processes were carried out with gentle agitation at room temperature. After rinsing with $1\times$ running buffer (25 mM Tris, 192 mM glycine and 0.1% w/v SDS), the equilibrated strips were aligned on top of 12% SDS-PAGE gels and were sealed with overlay agarose (Bio-Rad, USA). For the second dimension protein separation, the SDS-PAGE of two gels was performed in the PROTEAN II xi Cell (Bio-Rad, USA) containing 1X running buffer at 50 mA for 30 min, followed by 70 mA until the bromophenol blue dye front reached the bottom of the gel.

Silver staining of gel and image analysis

The gels were subjected to silver staining using the PlusOne Silver Staining Kit (GE Healthcare, USA) according to modified protocols that are compatible with mass spectrometry (Yan et al. 2000). The staining was performed according to the manufacturer's instruction except that glutaraldehyde was omitted during the sensitising step. The silver-stained gels were scanned using the GS-800 calibrated densitometer (Bio-Rad, USA). The digitised proteome maps of untreated control and MNQ-treated cells were analysed and compared in triplicates using the PDQuest Basic 2-D Analysis Software, version 8.0 (Bio-Rad, USA). The protein spots were automatically detected and manually checked for undetected or incorrectly detected spots. Total

density in gel images was used as the normalization method. Quantitative analysis was performed using Student's *t*-test between the proteome maps of untreated control and MNQ-treated sample. Protein spots with more than 1.5 fold change ($p < 0.05$) were selected for in-gel tryptic digestion. Molecular masses and isoelectric points (pIs) of these spots were estimated using the 2D SDS-PAGE Standards (Bio-Rad, USA).

In-gel tryptic digestion

Selected protein spots were manually excised and cut into smaller pieces using clean scalpel blades and forceps. The gel pieces were then destained as described by Gharahdaghi et al. (1999) with slight modification. Briefly, the gel pieces were incubated in the freshly prepared 1:1 solution of 30 mM potassium ferricyanide (Merck, Darmstadt, Germany) and 100 mM sodium thiosulphate (Sigma, USA) with gentle shaking for 15 min at room temperature until the brownish silver colour disappeared. After that, the gel pieces were gently shaken in washing solution (25 mM ammonium bicarbonate (NH_4HCO_3) in 50% acetonitrile (ACN)) for 10 min, thrice. Further reduction and alkylation as well as the enzymatic digestion were carried out based on the manufacturer's instruction of the In-Gel Tryptic Digestion Kit (Pierce, Rockford, IL, USA). The resultant digestion mixture was subjected to sample clean-up using Pierce C18 Spin Column (Pierce, USA) and the eluted peptide sample was completely dried in a vacuum centrifuge (Eppendorf, Germany).

MALDI-TOF/TOF-MS

The dried peptide samples were first dissolved in 50% ACN/0.1% TFA and were spotted on a 384-well target plate. The reconstituted peptide samples were then crystallised in a total volume of 1 μl with a saturated matrix solution of α -cyano-4-hydroxycinnamic acid (at final concentration of 5 mg/ml) in 50% ACN/0.1% TFA. The peptide samples were analysed on the 4800 Proteomics Analyzer MALDI-TOF/TOF (Applied Biosystems, USA). The MS data of the sample was automatically acquired in the reflectron mode and common trypsin and keratin peaks were excluded. Ten most intense ions from peptide mass fingerprinting (PMF) data were automatically selected for further MS/MS fragmentation and analysis. The collision energy of the MS system was set at 1 kV and nitrogen was used as the collision gas.

Protein identification for all the samples was performed by submitting both MS and MS/MS data to the MASCOT v 2.1 database search engine (Matrix Science Ltd., London, UK). The database searching was conducted using NCBI nr protein database using the following parameter settings: mass error tolerance of 200 ppm for MS data and 0.4 Da for MS/MS data; carbamidomethylation of cysteine was selected as fixed modification whereas the oxidation of methionine was set as variable modification. For data interpretation purpose, GPS Explorer Software v 4.9 (Applied Biosystems, USA) was used for the further data analysis. Protein scores greater than 83 were considered to be statistically significant ($p < 0.05$). The identities of the proteins were reported if the statistically significant protein scores were achieved and the corresponding gel spot positions matched the theoretical molecular weight and isoelectric point values. The information on the biological functions of the identified proteins was obtained from the Uniprot website (<http://www.uniprot.org/>) and the available scientific literature (<http://www.pubmed.com/>).

Western blotting

Total protein extracts were obtained by lysing the cells in 1X RIPA buffer (Millipore, CA, USA) supplemented with 0.01% v/v SDS and 1% v/v Halt Protease and Phosphatase Inhibitor Cocktail (Pierce, USA) for 15 min at 4 °C. The concentrations of the obtained protein extracts were quantified using BCA Protein Assay (Pierce, USA).

Equivalent amounts of protein samples (20 $\mu\text{g}/\text{lane}$) were resolved using 10% or 12% SDS-PAGE to visualise laminin-binding protein (RPSA), E-cadherin, total and phospho-ERK1/2 (p-ERK1/2), as well as total and phospho-NF- κ B p65 (p-NF- κ B p65). For the detection of protein S100-A4 (S100A4), 12% Tris-Tricine SDS-PAGE was used instead to separate the protein samples. Resolved proteins were electroblotted onto the low fluorescent Hybond LFP 0.2 μm PVDF membranes (GE Healthcare, USA), which were then incubated for 1 h at room temperature in Odyssey Blocking Buffer (LI-COR Biosciences, USA) or in 1% BSA in 1X TBS for the detection of phosphoproteins.

The blocked membranes were incubated overnight at 4 °C with primary antibodies against S100A4 (1:2000), RPSA (1:1000), E-cadherin (1:1000), total and p-ERK1/2 (both 1:2000) as well as total and p-NF- κ B p65 (both 1:1000). After that, the membranes were washed with 1X TBST four times with 5 min per wash and incubated for 1 h with appropriate IRDye 680RD secondary antibody (1:5000) at room temperature in dark condition. Following the completion of secondary antibody probing, the membranes were washed four times with 1X TBST and two times with 1X TBS with 5 min per wash. The near infrared (NIR)-based fluorescent detection of the immunoreactive bands was achieved using the UVP Biospectrum® Imaging System (UVP, LLC, CA, USA).

In order to check for equal protein loading, the membranes were stripped using the Restore Fluorescent Western Blot Stripping Buffer (Pierce, USA) and were reprobed with primary antibody against β -actin (1:4000) which served as the loading control. The detection of the β -actin signal was performed as described above. Densitometric analysis was performed by using ImageJ 1.48v software (National Institutes of Health, USA). All the primary antibodies were prepared at their respective optimized dilution ratios in Odyssey Blocking Buffer containing 0.1% Tween-20 with the exception of p-ERK1/2 and p-NF- κ B p65 which were prepared in 1% BSA in 1X TBST instead. The secondary antibody solution was prepared by diluting the antibody at 1:5000 in either Odyssey Blocking Buffer or 1% BSA in 1X TBS, of which both were supplemented with 0.1% Tween-20 and 0.01% SDS.

Statistical analysis

All experiments were performed in triplicates independently. The difference between the treated samples and the untreated control was analysed by Student's *t*-test in which $p < 0.05$ was considered as significant.

Results

Differential proteomic expression in MNQ-treated and untreated MDA-MB-231 cells

Based on our previous findings, MNQ was able to exert anti-metastatic effects against MDA-MB-231 cells (Liew et al. 2014). In this study, a comparative proteomic approach was employed to facilitate the identification of target-specific proteins whose modulation could be crucial in mediating the anti-metastatic effects of MNQ. MDA-MB-231 cells were treated with either DMSO (control) or MNQ at 7.5 μM for 24 h. The extracted whole cell lysates of both samples were subjected to 2DGE in which more than 1000 protein spots were able to be consistently resolved and detected for both control and treated samples. The gels were digitally imaged and quantitatively analysed using PDQuest Basic 2-D Analysis Software. Protein spots that displayed significant differential expression (> 1.5 fold change, $p < 0.05$) upon MNQ treatment were excised and processed for MALDI-TOF/TOF-MS analysis. In total, 26 differentially expressed proteins were successfully identified using MALDI-TOF/TOF-MS with significant protein score. Among these, 19 were downregulated (Spots 1–19) and seven were upregulated (Spots 20–26). Representative silver-stained 2D proteome maps of both control and MNQ-treated cells from three

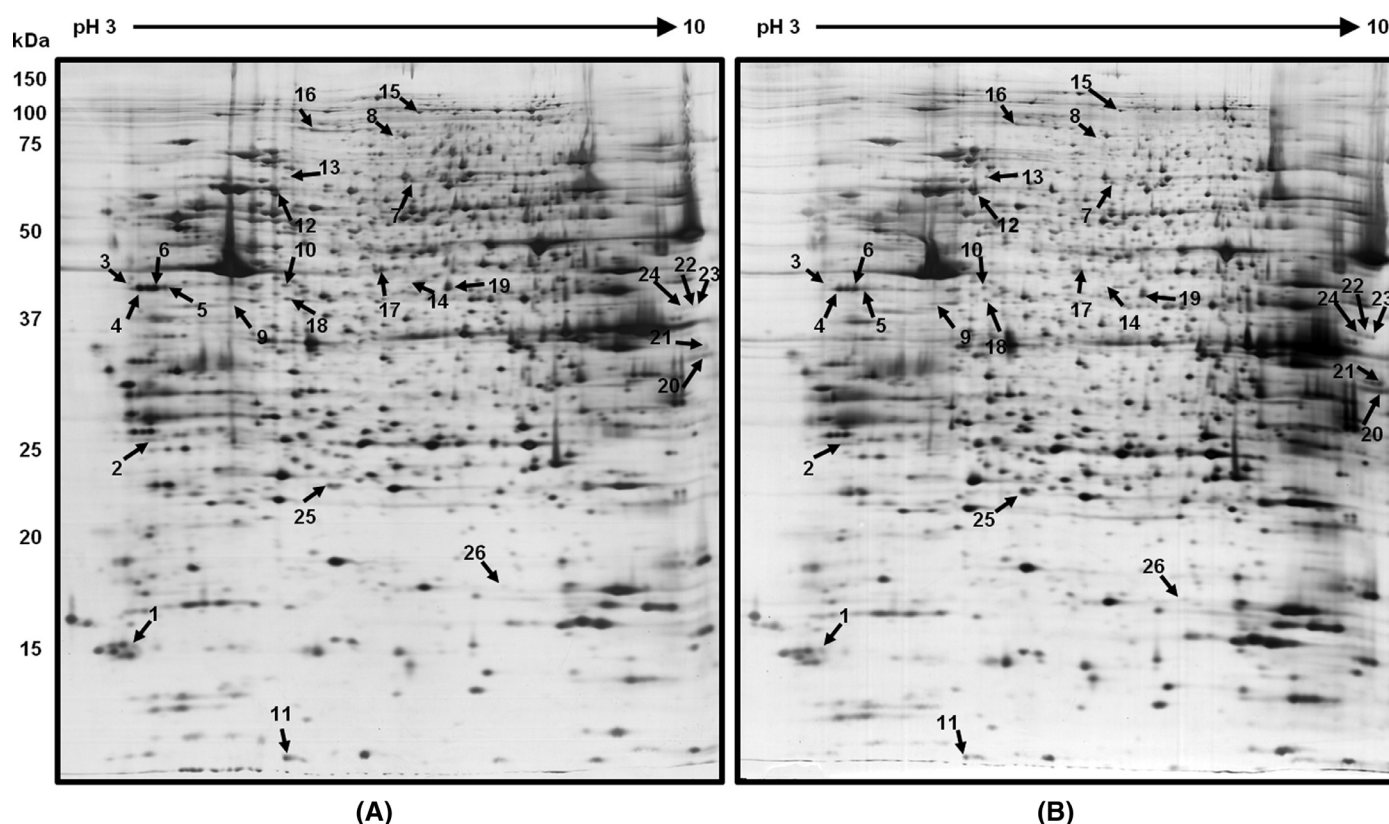


Fig. 2. Representative images of the 2D proteome maps of (A) untreated MDA-MB-231 cells and (B) MDA-MB-231 cells treated with 7.5 μ M MNQ for 24 h. 2DGE experiments were performed as described in the Materials and methods section. The arrowed spots indicate the differentially expressed proteins (>1.5 fold change, $p < 0.05$) and the numbers stated correspond to the spot numbers in Table 1.

independent experiments are shown in Fig. 2A and B and the identified proteins are marked with arrows.

Through bioinformatic analysis, the biological functions of these 26 differentially expressed proteins were obtained and are shown in Table 1. The proteins affected by MNQ are of diverse functions and profiles in which some of these proteins have more than one function. The expression of proteins associated with cytoskeleton and its regulations namely myosin light polypeptide 6 isoform 1 (MYL6), p27BBP protein (eIF6), dihydropyrimidinase-related protein 2 isoform 2 (DPYSL2), caldesmon (CALD1), T-complex protein 1 subunit theta isoform 1 (CCT8) and Chain A, crystal structure of AlixAIP1 (PDCD6IP) were all downregulated by MNQ. Furthermore, significant decrease in the expression level was also observed in a variety of proteins involved in ribosome processing (laminin receptor-like protein LAMRL5 (LAMR1P15)), membrane trafficking (copine-1 (CPNE1)), immune response (MHC Class I antigen (HLA-A)), DNA methylation (S-adenosylhomocysteine hydrolase (AHCY)), transcription and DNA repair (nuclear factor IV (XRCC5)), nucleotide metabolism (3'-(2'),5'-bisphosphate nucleotidase 1 (BPNT1)) as well as apoptosis and cell cycle (PDCD6IP).

In contrast, the expression of proteins linked with mRNA processing were all upregulated by MNQ with the exception of poly(rC)-binding protein 3 isoform 2 (PCBP3). These proteins include ELAV-like protein 1 (ELAVL1), heterogeneous nuclear ribonucleoprotein A1 isoforms a and b (HNRNPA1) and heterogeneous nuclear ribonucleoproteins A2/B1 isoforms A2 and B1 (HNRNPA2B1). Moreover, the level of protein DJ-1 (PARK7) which mediates oxidative stress response was found to be upregulated as well. On the other hand, proteins which regulate the protein modification processes such as folding, sumoylation, ubiquitination and phosphatase activity also displayed significant alteration in the expression level upon intervention with MNQ. The levels of SUMO-activating enzyme subunit 1 isoform a (SAE1),

COP9 complex subunit 4 (COPS4) and CCT8 were lowered whereas the level of low molecular weight phosphotyrosine protein phosphatase isoform c (ACP1) was increased. It is noteworthy that MNQ significantly modulated the expressions of proteins that are known to drive metastatic progression. Protein S100-A4 (S100A4) which promotes epithelial-to-mesenchymal transition (EMT) was found to be down-regulated. Laminin-binding protein (RPSA) which plays a role in the cell adhesion to basement membrane was similarly downregulated.

Confirmation of differential expression of S100A4 and RPSA

Both S100A4 and RPSA are known to promote metastasis given that there are positive correlations between increased expressions of these proteins and the metastatic potential of tumour cells (Garrett et al. 2006; Nelson et al. 2008). Due to their high relevance to tumour metastasis, further investigation was made on both S100A4 and RPSA in this study. Western blotting was employed to confirm the alterations of S100A4 and RPSA expressions upon MNQ treatment. The results obtained from the Western blot analysis were consistent with the findings from the proteomic analysis (Fig. 3A and B). The expression levels of S100A4 and RPSA were decreased in MDA-MB-231 cells after MNQ treatment (Fig. 3C and D). Additionally, MNQ at 7.5 μ M was also shown to reduce S100A4 and RPSA expressions in a time-dependent manner (Fig. 4A and B). Thus, the inhibitory effects of MNQ on the expression levels of S100A4 and RPSA in MDA-MB-231 cells were confirmed.

Activation statuses of ERK1/2 and NF- κ B

ERK1/2 signalling pathway has been shown to be involved in governing the expressions of both S100A4 (Chen et al. 2007; Zhang et al. 2007) and RPSA (Liu et al. 2010a, 2010b). S100A4 itself is also

Table 1The differentially expressed proteins in MDA-MB-231 cells upon treatment with MNQ at 7.5 μ M for 24 h.

Spot ^a	Accession no. ^b	Protein identity ^b	MW (kDa)/pI ^b	Score ^c	Fold change ^d	Biological functions
1	gi 17986258	Myosin light polypeptide 6 isoform 1 (MYL6)	17.09/4.56	120	–2.68	Cytoskeleton
2	gi 13785574	p27BBP protein (eIF6)	26.85/4.56	188	–2.74	Cytoskeleton-related, protein synthesis and ribosome processing
3	gi 14583014	Laminin receptor-like protein LAMRL5 (LAMR1P15)	33.09/4.84	193	–2.94	Ribosome processing
4	gi 34234	Laminin-binding protein (RPSA)	31.89/4.84	577	–2.69	Cell adhesion to basement membrane, ribosome processing
5	gi 34234	Laminin-binding protein (RPSA)	31.89/4.84	240	–2.99	
6	gi 34234	Laminin-binding protein (RPSA)	31.89/4.84	604	–2.29	
7	gi 4503377	Dihydropyrimidinase-related protein 2 isoform 2 (DPYSL2)	62.71/5.95	640	–2.07	Cytoskeletal organisation, signal transduction
8	gi 179830	Caldesmon (CALD1)	62.72/6.18	130	–4.11	Cytoskeleton
9	gi 4885585	SUMO-activating enzyme subunit 1 isoform a (SAE1)	38.88/5.17	97	–4.11	Protein modification, protein sumoylation
10	gi 5410300	COP9 complex subunit 4 (COPS4)	46.45/5.57	133	–1.89	Regulation of ubiquitination
11	gi 4506765	Protein S100-A4 (S100A4)	11.95/5.85	94	–4.03	Epithelial to mesenchymal transition, NF- κ B activation
12	gi 48762932	T-complex protein 1 subunit theta isoform 1 (CCT8)	60.15/5.18	395	–2.74	Chaperone, protein folding, cytoskeleton-related
13	gi 48145697	Copine-1 (CPNE1)	59.68/5.52	296	–2.19	Membrane trafficking
14	gi 150251581	MHC class I antigen (HLA-A)	36.32/5.83	250	–2.51	Immune response
15	gi 145580011	Chain A, crystal structure of AlixAIP1 (PDCD6IP)	79.40/5.66	233	–2.09	Protein transport, apoptosis and cell cycle, cytoskeleton-related
16	gi 35038	Nuclear factor IV (XRCC5)	71.65/5.71	165	–5.08	Transcription, DNA repair
17	gi 178277	S-Adenosylhomocysteine hydrolase (AHCY)	48.25/6.03	286	–2.08	DNA methylation
18	gi 116812595	3'(2'),5'-Bisphosphate nucleotidase 1 (BPNT1)	33.71/5.46	135	–2.63	Nucleotide metabolism
19	gi 194306633	Poly(rC)-binding protein 3 isoform 2 (PCBP3)	37.08/6.72	91	–2.05	mRNA processing
20	gi 38201714	ELAV-like protein 1 (ELAVL1)	36.24/9.23	218	+3.10	
21	gi 4504445	Heterogeneous nuclear ribonucleoprotein A1 isoform a (HNRNPA1)	34.29/9.27	330	+4.65	
22	gi 14043070	Heterogeneous nuclear ribonucleoprotein A1 isoform b (HNRNPA1)	38.84/9.17	360	+19.98	
23	gi 14043072	Heterogeneous nuclear ribonucleoproteins A2/B1 isoform B1 (HNRNPA2B1)	37.46/8.97	235	+13.02	
24	gi 4504447	Heterogeneous nuclear ribonucleoproteins A2/B1 isoform A2 (HNRNPA2B1)	36.04/8.67	414	+13.85	
25	gi 31543380	Protein DJ-1 (PARK7)	20.05/6.33	188	+2.69	Oxidative stress response
26	gi 4757714	Low molecular weight phosphotyrosine protein phosphatase isoform c (ACP1)	18.49/6.30	163	+3.72	Protein tyrosine phosphatase activity

^a Spot number corresponds to those included in the representative 2D proteome maps (Fig. 2).^b Accession number, protein identity, theoretical molecular weight (MW) and isoelectric point (pI) were obtained from NCBI database.^c MASCOT protein score >83 were considered to be statistically significant ($p < 0.05$).^d (–): downregulated; (+): upregulated.

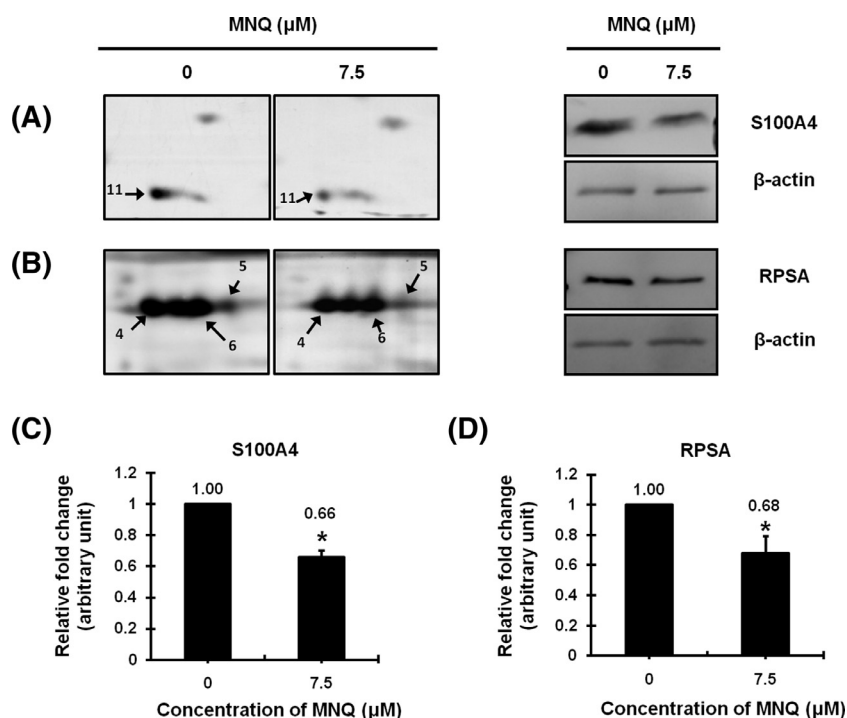


Fig. 3. Confirmation of differential expression of S100A4 and RPSA in MDA-MB-231 cells upon treatment with MNQ. Whole cell lysate of untreated and 7.5 μM MNQ-treated MDA-MB-231 cells for 24 h were analysed using Western blotting. The zoomed views of 2D gel regions for (A) S100A4 (Spot 11) and (B) RPSA (Spots 4–6) are shown in the left panels whereas their corresponding Western blot analyses are shown in the right panels. Representative 2D gel and blot images for three independent experiments are shown. Western blot densitometric analysis of (C) S100A4 and (D) RPSA expressions was performed using ImageJ software. β-Actin was used as the loading control in which S100A4 and RPSA expressions were normalised to the corresponding β-actin levels. Fold change values relative to the untreated control were shown in the bar charts. Data are expressed as means ± standard deviations for three independent experiments. Statistical analysis was performed using Student's *t*-test. **p* < 0.05; significant difference from the untreated control.

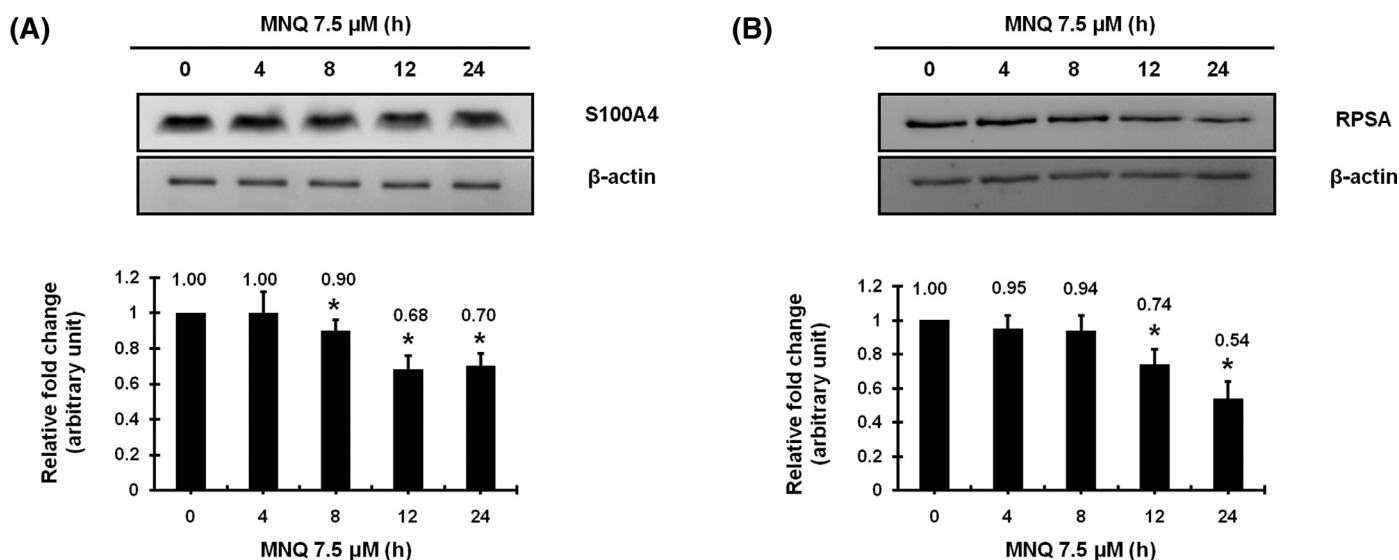


Fig. 4. Time-course study for the expressions of (A) S100A4 and (B) RPSA in MDA-MB-231 cells upon treatment with 7.5 μM MNQ for 0, 4, 8, 12 and 24 h. Representative Western blot images for three independent experiments are shown. Densitometric analysis was performed using ImageJ software. β-actin was used as the loading control in which S100A4 and RPSA expressions were normalised to the corresponding β-actin levels. Fold change values relative to the untreated control were shown in the bar charts. Data are expressed as means ± standard deviations for three independent experiments. Statistical analysis was performed using Student's *t*-test. **p* < 0.05; significant difference from the untreated control.

known to play an important role in the regulation of NF-κB signalling pathway (Grotterød et al. 2010). Hence, we speculated that MNQ may affect the activations of both pathways. In order to identify the regulatory effects of MNQ on ERK1/2 and NF-κB activations in MDA-MB-231 cells, Western blot analysis using antibodies specific to both phosphorylated and total ERK1/2 and NF-κB p65 was performed. The activation status of ERK1/2 which correlates positively

with ERK1/2 phosphorylation, was observed to be repressed by MNQ at 7.5 μM in the time course study as highlighted by the decreasing level of p-ERK1/2 across 48 h of treatment period (Fig. 5A). The level of total ERK1/2 in contrast, remained unchanged which indicates that the expression of ERK1/2 was not affected by MNQ. p-ERK1/2 level was initially elevated at earlier time points and the reduction of the level of p-ERK1/2 began to become apparent from 24th hour

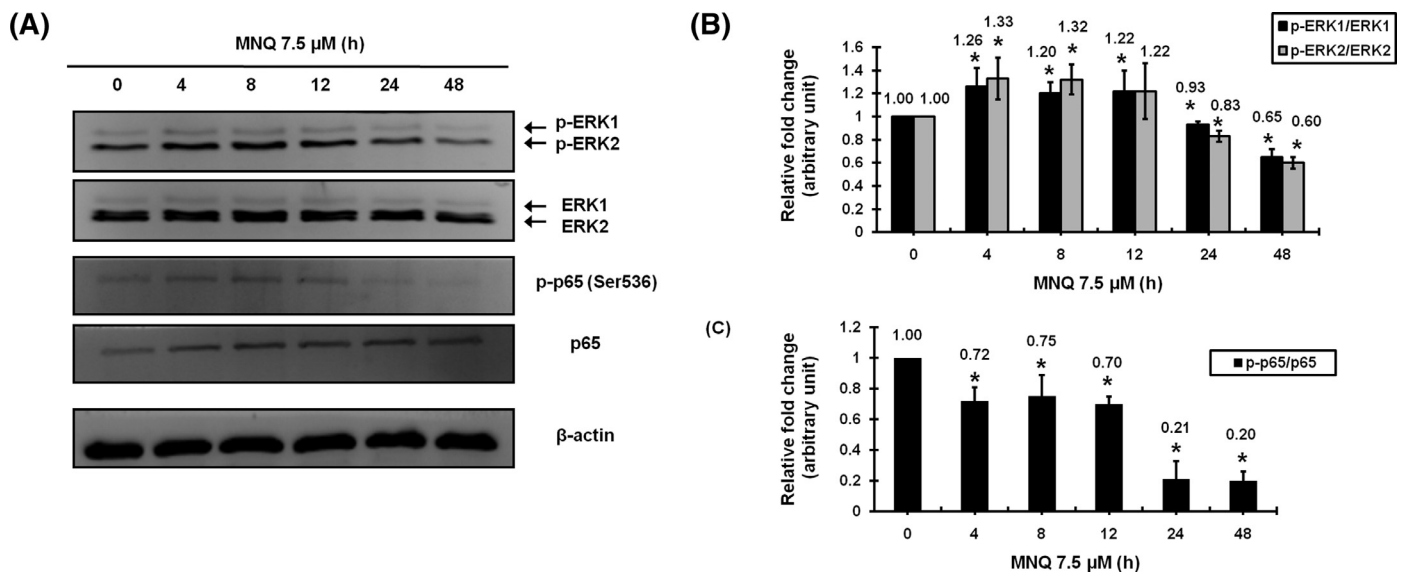


Fig. 5. The effects of MNQ on the activation statuses of ERK1/2 and NF- κ B in MDA-MB-231 cells. The whole cell lysates of MDA-MB-231 cells treated with 7.5 μ M MNQ for different time intervals were subjected to Western blot analysis. (A) Representative blot images depicting the levels of p-ERK1/2, total ERK1/2, p-NF- κ B p65, total NF- κ B p65 and the loading control, β -actin for three independent experiments are shown. Densitometric analysis of (B) ERK1/2 and (C) NF- κ B activations was performed using ImageJ software whereby the levels of phosphoproteins were normalised to the corresponding levels of their total proteins. Fold change values relative to the untreated control were shown in the bar charts. Data are expressed as means $\pm$ standard deviations for three independent experiments. Statistical analysis was performed using Student's *t*-test. **p* < 0.05; significant difference from the untreated control.

onwards. The lowest p-ERK1/2 level was observed at the 48th hour in which MNQ was shown to reduce the levels of p-ERK1 and p-ERK2 by 35% and 40% respectively as compared to the untreated control (Fig. 5B).

On the other hand, MNQ at 7.5 μ M was also found to suppress the phosphorylation of NF- κ B p65 at Ser536 (Fig. 5A). A strong reduction in the level of p-NF- κ B p65 was seen at 24- and 48-h treatment intervals in this time course study. Conversely, there was an increase in the total p65 level at the fourth hour after which the level of total p65 remained fairly constant. Nonetheless, the possibility of unequal loading was excluded as reprobing with anti- β -actin antibody indicated that the levels of β -actin loading control were relatively equal for all time points (Fig. 5A). Based on the densitometric analysis, remarkable suppression of p65 phosphorylation by MNQ took place at 24- and 48-h treatment intervals (Fig. 5C). MNQ reduced the level of p-p65 by approximately 80% relative to the untreated control at both time points. Since NF- κ B p65 phosphorylation is required for optimal NF- κ B activation (Huang et al. 2010), this observation lends credence to the potential of MNQ in inhibiting the NF- κ B pathway.

Expression of E-cadherin

The loss of the epithelial marker, E-cadherin during EMT progression is intimately connected to high S100A4 expression (Yonemura et al. 2000; Moriyama-Kita et al. 2005) and aberrant NF- κ B activation (Chua et al. 2007). Western blotting was used to assess the effect of MNQ on the expression of E-cadherin and the result of the time course study is shown in Fig. 6. MDA-MB-231 cells are known to be E-cadherin-deficient and have extremely low basal expression of E-cadherin (Lombaerts et al. 2006) as evidenced at 0 h. However, MNQ at 7.5 μ M was observed to stimulate the expression of E-cadherin in MDA-MB-231 cells. The highest induction took place at the 48th hour in which E-cadherin expression was increased up to 3.19 fold as compared to the untreated control. The induction of E-cadherin expression by MNQ in the E-cadherin-deficient MDA-MB-231 cells suggests that MNQ could have the potential to inhibit EMT by restoring the epithelial characteristic in the cells.

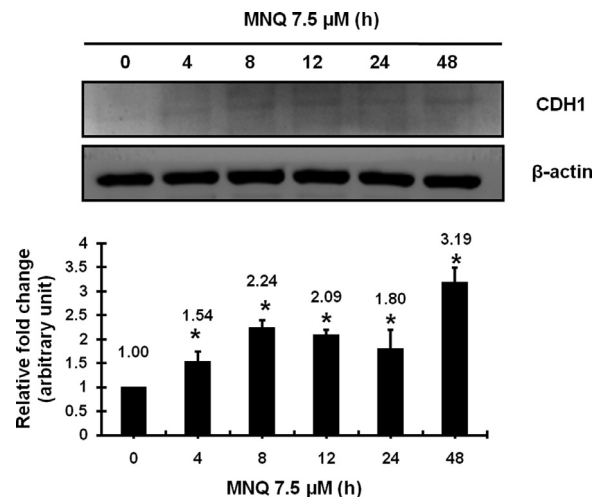


Fig. 6. The effect of MNQ towards the expression of E-cadherin (CDH1) in MDA-MB-231 cells. The whole cell lysates of MDA-MB-231 cells treated with 7.5 μ M MNQ for different time intervals were subjected to Western blot analysis. Representative blot images for three independent experiments are shown. Densitometric analysis was performed using ImageJ software. β -Actin was used as the loading control in which E-cadherin expressions were normalised to the corresponding β -actin levels. Fold change values relative to the untreated control were shown in the bar charts. Data are expressed as means $\pm$ standard deviations for three independent experiments. Statistical analysis was performed using Student's *t*-test. **p* < 0.05; significant difference from the untreated control.

Discussion

We have previously demonstrated that MNQ inhibited the invasion, migration and proteolytic activity of MMP-9 of MDA-MB-231 cell line at low cytotoxic concentrations (≤ 7.5 μ M) (Liew et al. 2014). Yet, little is known about its specific mechanism of action. In this study, we employed the 2DGE and MALDI-TOF/TOF-MS to identify the differentially expressed proteins in this cell line upon exposure to MNQ at 7.5 μ M. These proteins could potentially represent the target-specific proteins of MNQ, and thus, shed light on the molecular mechanism

of its anti-metastatic effects. In total, we have identified 26 proteins whose expressions were significantly changed by MNQ intervention.

Investigation at the proteome level revealed significant downregulation of a large number of cytoskeletal-related proteins (MYL6, eIF6, CCT8, DPYSL2, PDCD6IP, and CALD1). It is widely accepted that cancer cells undergo dramatic cytoskeletal reorganisation to acquire migratory and invasive properties (Yilmaz and Christofori 2009). MYL6 is involved in structural stability and motor function of nonmuscle myosin II whereby this myosin complex mediates cell contractility, adhesion and migration (Hernandez et al. 2007). eIF6 is known to interact with $\beta 4$ integrin and intermediate filament cytoskeleton and these interactions could support its roles as the downstream effector of Notch pathway to enhance metastasis (Benelli et al. 2012). In terms of CCT8, it is one of the subunits in the molecular chaperone TCP-1 ring complex that assists the assembly, folding and polymerisation of actin (Brackley et al. 2010). DPYSL2 has been shown to promote microtubule assembly and stabilisation (Lin et al. 2011) whereas PDCD6IP was found to be critical in the formation of F-actin, stress fibres and lamellipodial membrane protrusions (Pan et al. 2006). On the other hand, the upregulation of CALD1, a protein that mediates actin-myosin interaction was reported to coincide with increased HeLa cell migration (Tseng et al. 2011). The downregulation of the above proteins could therefore lead to the blockade of cytoskeletal reorganisation which results in the impediment of the metastatic potential of MDA-MB-231 cells by MNQ.

Furthermore, the expressions of two well-known metastatic mediators, S100A4 and RPSA were notably downregulated by MNQ. The involvement of S100A4 in promoting metastasis is well-documented in several studies. Li and Bresnick (2006) reported that the binding of S100A4 to the nonmuscle myosin-IIA (NMIIA) enhanced the directional motility during chemotaxis whereby the high S100A4 expressing cells exhibited fewer side protrusions and more extensive forward protrusions towards the chemoattractant. The increased directional motility could be mediated by the regulation of NMIIA filaments turnover at the leading edge of cells by S100A4 (Li and Bresnick 2006). House et al. (2011) have also shown that S100A4-NMIIA interaction was vital for cancer cell invasion. In addition, S100A4 can promote the expression and activity of various MMPs leading to the enhanced proteolytic degradation of basement membrane barrier during metastasis (Elenjord et al. 2008). As for RPSA, this laminin receptor can promote metastasis by facilitating the cancer cell adhesion to the laminin component in the basement membrane (Nelson et al. 2008). Invading tumour cells often display augmented metastatic potential upon interacting with laminin (Givant-Horwitz et al. 2005). Ardini et al. (2002) reported that RPSA can remodel laminin substrate thereby increasing the laminin degradation mediated by the protease Cathepsin B. Furthermore, RPSA was found to be capable of directly interacting with actin and this interaction was critical for the migration of NIH 3T3 cell line (Ventecineque et al. 2011).

Since S100A4 and RPSA are consistently implicated in the manifestation of metastatic potential of various cancers cells (Garrett et al. 2006; Nelson et al. 2008), both proteins were further investigated in this study. The downregulation of the expressions of S100A4 and RPSA by MNQ shown in the proteomic analysis were subsequently confirmed using Western blot analysis (Fig. 3A–D). Reduction in the protein levels of both S100A4 and RPSA in a time-dependent manner was also observed (Fig. 4A and B). Therefore, it is conceivable that MNQ-mediated downregulation of S100A4 and RPSA led to the suppression of the metastatic potential of MDA-MB-231 cells as observed in our previous findings (Liew et al. 2014).

Moreover, we examined the effect of MNQ towards the activation of ERK1/2 signalling pathway in MDA-MB-231 cells. As shown in Fig. 5A and B, MNQ reduced the phosphorylation of ERK1/2 from the 24th hour and the decrease in the p-ERK1/2 level continued right up to the 48th hour of treatment period. This indicates that the activation of ERK1/2 signalling pathway in these cells can be inhibited after

a prolonged exposure to MNQ. As this pathway has been reported to positively regulate the expressions of S100A4 (Chen et al. 2007; Zhang et al. 2007) and RPSA (Liu et al. 2010a, 2010b), the observed reduction of the cellular levels of S100A4 and RPSA could be attributed to the inhibition of ERK1/2 activation in MDA-MB-231 cells by MNQ. It is also noteworthy that the dysregulation of ERK1/2 pathway itself plays a central role in metastasis. Seddighzadeh et al. (1999) reported that ERK1/2 signalling in MDA-MB-231 cells is tuned to a level that maintains high expression of urokinase plasminogen activator (uPa), another pro-metastatic protease that in turn contributes to the invasive behaviour of these cells. Huang et al. (2004) also pointed out that ERK1/2 signalling can actively mediate cell migration via phosphorylation of myosin light chain kinase, calpain, focal adhesion kinase (FAK) and paxillin in order to promote membrane protrusions and focal adhesion disassembly. As such, the observed inhibition of MDA-MB-231 cell invasion and migration by MNQ (Liew et al. 2014) could also be the result of inactivation of ERK1/2 signalling pathway.

Besides ERK1/2, we also investigated the effects of MNQ on NF- κ B signalling pathway in MDA-MB-231 cells. The present data showed that MNQ repressed the phosphorylation of p65, a major member in NF- κ B transcription factor family at Ser536 (Fig. 5A and C). Previously, p65 phosphorylation at Ser536 has been reported to enhance the nuclear translocation of p65, leading to stronger NF- κ B transcriptional activity (Huang et al. 2010). This indicates that MNQ could inhibit the activation of NF- κ B in MDA-MB-231 cells at least via suppression of p65 phosphorylation at Ser536. This observation also corroborates the reduction of ERK1/2 activation and S100A4 expression by MNQ observed in this study. Chen and Lin (2001) reported that the inhibition of ERK1/2 was able to attenuate the p65 transactivation. Similarly, Zhang et al. (2013) showed that the knockdown of S100A4 using RNA interference was able to inhibit p65 nuclear translocation. Furthermore, the data from the proteomic analysis may also offer further insight on the inactivation of NF- κ B signalling pathway by MNQ. SAE1, COPS4 and CPNE1 have been shown in several studies to participate in the ubiquitin-mediated proteosomal degradation of I κ B, an endogenous inhibitor that sequesters NF- κ B in the cytoplasm, which in turn activates NF- κ B (Fu et al. 2001; Tomsig et al. 2004; Aillet et al. 2012). The simultaneous downregulation of SAE1, COPS4 and CPNE1 by MNQ as shown in this study could therefore disrupt the proteosomal degradation of I κ B, thereby inhibiting the activation of NF- κ B. As NF- κ B activation can upregulate the expressions of various pro-metastatic genes (Basseres and Baldwin 2006), the reduction of the metastatic potential of MDA-MB-231 cells observed in our previous study (Liew et al. 2014) could well be ascribed to MNQ-mediated inhibition of NF- κ B signalling pathway.

E-cadherin is a vital metastatic suppressor that mediates cell–cell adhesion in order to maintain the epithelial integrity in the epithelial tissues. The loss of E-cadherin and its functions are frequently observed during EMT which leads to the acquisition of invasive phenotypes by the cancer cells (Pečina-Šlaus 2003). MDA-MB-231 cell line is known to be E-cadherin-deficient and has extremely low basal expression of E-cadherin (Lombaerts et al. 2006). In this study, we found that MNQ induced the expression of E-cadherin in these cells (Fig. 6). Although only relatively low expression was induced from the fourth hour of treatment period onwards, the E-cadherin level was in stark contrast to what was observed in the untreated control in which E-cadherin expression was barely detectable. Continuous induction of E-cadherin expression in MDA-MB-231 cells was observed throughout the 48-h treatment period. This result is in agreement with our earlier findings in which S100A4 expression and NF- κ B activation were inhibited by MNQ. Inverse relationship between S100A4 and E-cadherin expression was previously described in various tumour cell lines in which overexpression of S100A4 correlates with the occurrence of EMT and reduced E-cadherin expression (Yonemura et al. 2000; Moriyama-Kita et al. 2005). NF- κ B was also proven to mediate EMT via upregulation of EMT-associated transcription factors such as

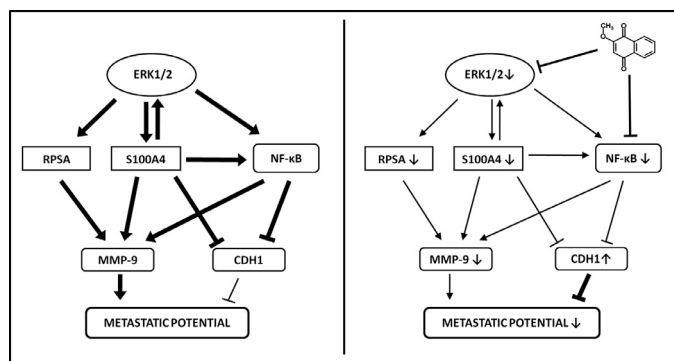


Fig. 7. The hypothetical signalling network involved in MNQ-mediated anti-metastatic effects. Pointed arrows represent upregulation or activation whereas blocked arrows represent downregulation or inhibition. Upon MNQ intervention (right panel), the activations of ERK1/2 and NF- κ B signalling pathway are inhibited. As a result, the activities or expressions of downstream proteins in the network are altered in contrast to the network prior to MNQ intervention (left panel). These subsequently lead to the suppression of metastatic potential of MDA-MB-231 cells. Growing or shrinking of size of the arrows represents the changes in the interaction between each protein that contribute to the alteration of the activity and expression of a particular protein.

Snail, Twist1, Zeb1 and Zeb2 which in turn transcriptionally repress E-cadherin (Chua et al. 2007; Julien et al. 2007; Li et al. 2012). Given all the above considerations, MNQ could stimulate the expression of E-cadherin in MDA-MB-231 cell line via inhibition of S100A4 expression and NF- κ B signalling pathways. This might contribute partly to the restoration of epithelial behaviour, leading to the reduced metastatic potential of MDA-MB-231 cells. However, further studies are required to prove the potential of MNQ as an EMT inhibitor.

It is also worth mentioning that MMP-9, whose proteolytic activity was inhibited by MNQ in our previous study (Liew et al. 2014), can be regulated by S100A4 and RPSA. Various studies have shown that siRNA-mediated silencing of S100A4 and RPSA can decrease the expression and activity of this pro-metastatic protease (Liu et al. 2010a, 2010b; Matsuura et al. 2010; Wang et al. 2013b). Overexpression of S100A4 was also demonstrated to upregulate the expression and activity of MMP-9 which led to the enhanced metastatic potential of tumour cells (Matsuura et al. 2010). Apart from that, both ERK1/2 and NF- κ B signalling pathways are known to regulate MMP-9 expression and activity. ERK1/2 positively regulates the expression of uPa (Seddighzadeh et al. 1999), a protease that is involved in the conversion of the inactive zymogen pro-MMP-9 into the enzymatically active MMP-9 (Ramos-DeSimone 1999). NF- κ B, on the other hand can transcriptionally regulate the gene expression of MMP-9 due to the presence of NF- κ B binding site in the human MMP-9 gene promoter (Yan and Boyd 2007). Taken together, the findings that MNQ inhibited the expressions of S100A4 and RPSA while simultaneously decreasing the activations of ERK1/2 and NF- κ B support our previous findings whereby MNQ suppressed the metastatic potential of MDA-MB-231 cells via the downregulation of MMP-9 activity (Liew et al. 2014). Based on the findings in this study and together with those from our own previous report (Liew et al. 2014), we propose a hypothetical model (Fig. 7) depicting the signalling network that could be affected by MNQ in order to potentiate its anti-metastatic effects against MDA-MB-231 cells. In this hypothetical model (Fig. 7), MNQ suppresses the metastatic potential of MDA-MB-231 cell line via the downregulation of S100A4 and RPSA expressions as well as the perturbation of ERK1/2 and NF- κ B signalling pathways; leading to the induction of E-cadherin expression and inhibition of MMP-9 activity.

However, this model does not yet answer the question of how MNQ affects this signalling network. Other studies have demonstrated the capabilities of MNQ in triggering the generation of reactive oxygen species (ROS) such as hydrogen peroxide (H_2O_2) and superoxide anion ($O_2^{\cdot-}$), causing increased intracellular ROS level and oxidative

stress (Tan 2011; Wang and Lin 2012). The notion of MNQ triggering ROS generation and oxidative stress in MDA-MB-231 cells can be supported by the upregulation of PARK7, ELAVL1, HNRNPA1, and HNRNPA2B1 observed in our proteomic study. PARK7 was reported to facilitate the cellular defence against ROS-induced oxidative stress (Im et al. 2012). ELAVL1, HNRNPA1 and HNRNPA2B1 are mRNA processing proteins that have been shown to participate in protecting target mRNAs related to stress response functions from degradation during oxidative catastrophe (Guil et al. 2006; Amadio et al. 2008; McDonald et al. 2011). Thus, the observed upregulation of these proteins may actually indicate the presence of elevated ROS level in MNQ-treated MDA-MB-231 cells. These proteins could be involved in the cellular response to adapt and to survive the possible oxidative insult caused by the exposure to MNQ. The potential ROS generation by MNQ can evoke oxidative modifications that alter the conformations of ERK1/2 and subsequently reduce their ability to be phosphorylated by upstream kinase activators such as MEK1/2 (Galli et al. 2008; Luanpitpong et al. 2012). Likewise, ROS that were possibly generated can oxidatively inactivate kinases that phosphorylate NF- κ B p65 at Ser536 such as I κ B kinase and Akt, leading to the repression of NF- κ B activation (Martin et al. 2002; Reynaert et al. 2006; Huang et al. 2010).

It is worth mentioning that ROS generation and modulation of intracellular signalling pathways have also been associated with the anti-metastatic effects of other 1,4-naphthoquinone derivatives. Kim et al. (2014) suggested that the suppression of an ovarian cancer cell (OVCAR-3) invasion via FAK and integrin- β 1 inhibition by menadione could be attributed to its promoting effect on ROS formation. In a study by Chen et al. (2014), shikonin was shown to impede the invasion and migration of PC-3 and DU145 prostate cancer cells via ROS-dependent activations of JNK1/2, p38 and ERK1/2 pathways. While the effect of shikonin on ERK1/2 pathway as reported by Chen et al. (2014) differ from that of MNQ, shikonin's regulation of ERK1/2 activation appears to be cell-type dependent. Wang et al. (2013a) showed that the inhibition of A549 lung cancer cell invasion and migration by shikonin required blockade of ERK1/2 activation. Similarly, Yang et al. (2013) found that the anti-metastatic effects of shikonin against K1 and FTC133 thyroid cancer cells involved the attenuation of ERK1/2 phosphorylation. Plumbagin, on the other hand was shown to inhibit the activation of ERK1/2 and DNA-binding activities of NF- κ B and AP-1, which subsequently reduced MMP-2 and uPa expressions in A549 cell (Shieh et al. 2010). NF- κ B inhibition was also described in the anti-metastatic activities of plumbagin against MDA-MB-231 and HepG2 liver cancer cells as well (Shih et al. 2009; Manu et al. 2011). As for thymoquinone, the reported anti-metastatic mechanisms of this compound include the inhibition of ERK1/2, FAK and NF- κ B pathways (Wu et al. 2011; Kolli-Bouhafs et al. 2012). Overall, all the reports above serve to strengthen our findings that MNQ could perturb the intracellular signalling such as ERK1/2 and NF- κ B pathways in MDA-MB-231 cells in order to potentiate its own anti-metastatic effects.

In conclusion, we have shown from our comparative proteomic analysis that MNQ significantly induced differential protein expressions in MDA-MB-231 cells. The downregulation of two well-known metastatic mediators, S100A4 and RPSA afforded an early insight on its previously reported anti-metastatic effect against MDA-MB-231 cell line. Further investigation on the signalling pathways that converge on S100A4 and RPSA revealed that MNQ disrupted the activations of ERK1/2 and NF- κ B pathways and induced E-cadherin expression in MDA-MB-231 cells. The previously reported MMP-9 inhibition by MNQ could also be attributed to the reduced S100A4 and RPSA expressions and the disruption of ERK1/2 and NF- κ B signalling pathways. It is possible that MNQ triggered ROS-driven oxidative stress to initiate perturbation of intracellular signalling network that culminates in the inhibition of the metastatic potential of MDA-MB-231 cells. This postulation however requires further studies. On the whole, the findings in this study shed light on the molecular mechanism of MNQ-mediated anti-metastatic effects. These findings could be useful

in the development of MNQ and other related 1,4-naphthoquinones as potent anti-metastatic agents for future clinical use.

Conflict of interest

The authors declare that they have no conflict of interest.

Acknowledgements

This work was supported by Ministry of Higher Education of Malaysia under the Fundamental Research Grant Scheme (FRGS/1/2011/SKK/TAYLOR/02/3) and Taylor's University, Malaysia through Taylor's Research Grant Scheme & Taylor's Fellowship Program.

References

- Aillet, F., Lopitz-Otsoa, F., Egaña, I., Hjerpe, R., Fraser, P., Hay, R.T., Rodriguez, M.S., Lang, V., 2012. Heterologous SUMO-2/3-ubiquitin chains optimize *IκBα* degradation and NF- κ B activity. *PLoS One* 7, e51672.
- Amadio, M., Scapagnini, G., Laforenza, U., Intrieri, M., Romeo, L., Govoni, S., Pascale, A., 2008. Post-transcriptional regulation of HSP70 expression following oxidative stress in SH-SY5Y cells: the potential involvement of the RNA-binding protein HuR. *Curr. Pharm. Des.* 14, 2651–2658.
- Ardini, E., Sporchia, B., Pollegioni, L., Mudugno, M., Ghirelli, C., Castiglioni, F., Tagliabue, E., Ménard, S., 2002. Identification of a novel function for 67-kDa laminin receptor: increase in laminin degradation rate and release of motility fragments. *Cancer Res.* 62, 1321–1325.
- Basseres, D.S., Baldwin, A.S., 2006. Nuclear factor- κ B and inhibitor of κ B kinase pathways in oncogenic initiation and progression. *Oncogene* 25, 6817–6830.
- Benelli, D., Cialfi, S., Pinzaglia, M., Talora, C., Londei, P., 2012. The translation factor eIF6 is a Notch-dependent regulator of cell migration and invasion. *PLoS One* 7, e32047.
- Brackley, K.I., Grantham, J., 2010. Subunits of the chaperonin CCT interact with F-actin and influence cell shape and cytoskeletal assembly. *Exp. Cell Res.* 316, 543–553.
- Chen, B.C., Lin, W.W., 2001. PKC- and ERK-dependent activation of I kappa B kinase by lipopolysaccharide in macrophages: enhancement by P2Y receptor-mediated CaMK activation. *Br. J. Pharmacol.* 134, 1055–1065.
- Chen, P.S., Wang, M.Y., Wu, S.N., Su, J.L., Hong, C.C., Chuang, S.E., Chen, M.W., Hua, K.T., Wu, Y.L., Cha, S.T., Babu, M.S., Chen, C.N., Lee, P.H., Chang, K.J., Kuo, M.L., 2007. CTGF enhances the motility of breast cancer cells via an integrin- α v β 3-ERK1/2-dependent S100A4-upregulated pathway. *J. Cell Sci.* 120, 2053–2065.
- Chen, Y., Zheng, L., Liu, J., Zhou, Z., Cao, X., Lv, X., Chen, F., 2014. Shikonin inhibits prostate cancer cells metastasis by reducing matrix metalloproteinase-2/-9 expression via AKT/mTOR and ROS/ERK1/2 pathways. *Int. Immunopharmacol.* 21, 447–455.
- Chua, H.L., Bhat-Nakshatri, P., Clare, S.E., Morimiya, A., Badve, S., Nakshatri, H., 2007. NF- κ B represses E-cadherin expression and enhances epithelial to mesenchymal transition of mammary epithelial cells: potential involvement of ZEB-1 and ZEB-2. *Oncogene* 26, 711–724.
- Ding, Z.S., Jiang, F.S., Chen, N.P., Lv, G.Y., Zhu, C.G., 2008. Isolation and identification of an anti-tumor component from leaves of *Impatiens balsamina*. *Molecule* 13, 220–229.
- Elenjor, R., Ijones, H., Sundkvist, E., Loennechen, T., Winberg, J.O., 2008. Dysregulation of matrix metalloproteinases and their tissue inhibitors by S100A4. *Connect. Tissue Res.* 49, 185–188.
- Fu, H., Reis, N., Lee, Y., Glickman, M.H., Vierstra, R.D., 2001. Subunit interaction maps for the regulatory particle of the 26S proteasome and the COP9 signalosome. *EMBO J.* 20, 7096–7107.
- Galli, S., Antico Arciuch, V.G., Poderoso, C., Converso, D.P., Zhou, Q., Bal de Kier, Joffé, E., Cadenas, E., Boczkowski, J., Carreras, M.C., Poderoso, J.J., 2008. Tumor cell phenotype is sustained by selective MAPK oxidation in mitochondria. *PLoS One* 3, e2379.
- Garrett, S.C., Varney, K.M., Weber, D.J., Bresnick, A.R., 2006. S100A4, a mediator of metastasis. *J. Biol. Chem.* 281, 677–680.
- Gharahdaghi, F., Weinberg, C.R., Meagher, C.A., Imai, B.S., Mische, S.M., 1999. Mass spectrometric identification of proteins from silver-stained polyacrylamide gel: a method for the removal of silver ions to enhance sensitivity. *Electrophoresis* 20, 601–605.
- Givant-Horwitz, V., Davidson, B., Reich, R., 2005. Laminin-induced signaling in tumor cells. *Cancer Lett.* 223, 1–10.
- Grotterød, I., Maelandsmo, G.M., Boye, K., 2010. Signal transduction mechanisms involved in S100A4-induced activation of the transcription factor NF- κ B. *BMC Cancer* 10, 241.
- Guil, S., Long, J.C., Cáceres, J.F., 2006. hnRNP A1 relocalization to the stress granules reflects a role in the stress response. *Mol. Cell Biol.* 26, 5744–5758.
- Guo, S., Zou, J., Wang, G., 2013. Advances in the proteomic discovery of novel therapeutic targets in cancer. *Drug Des. Devel. Ther.* 7, 1259–1271.
- Hernandez, O.M., Jones, M., Guzman, G., Szczesna-Cordary, D., 2007. Myosin essential light chain in health and disease. *Am. J. Physiol. Heart Circ. Physiol.* 292, H1643–H1654.
- House, R.P., Pozzuto, M., Patel, P., Dulyaninova, N.G., Li, Z.H., Zencheck, W.D., Vitolo, M.I., Weber, D.J., Bresnick, A.R., 2011. Two functional S100A4 monomers are necessary for regulating nonmuscle myosin-IIA and HCT116 cell invasion. *Biochemistry* 50, 6920–6932.
- Huang, C., Jacobson, K., Schaller, M.D., 2004. MAP kinase and cell migration. *J. Cell Sci.* 117, 4619–4628.
- Huang, B., Yang, X.D., Lamb, A., Chen, L.F., 2010. Posttranslational modifications of NF- κ B: another layer of regulation for NF- κ B signaling pathway. *Cell. Signal.* 22, 1282–1290.
- Im, J.Y., Lee, K.W., Woo, J.M., Junn, E., Mouradian, M.M., 2012. DJ-1 induces thioredoxin 1 expression through the Nrf2 pathway. *Hum. Mol. Genet.* 21, 3013–3024.
- Julien, S., Puig, I., Caretti, E., Bonaventure, J., Nelles, L., van Roy, F., Dargemont, C., Garcia de Herreros, A., Bellacosa, A., Larue, L., 2007. Activation of NF- κ B by Akt upregulates snail expression and induces epithelium mesenchyme transition. *Oncogene* 26, 7445–7456.
- Kraljevic, S., Sedick, M., Scott, M., Gehrig, P., Schlapbach, R., Pavelic, K., 2006. Casting light on molecular events underlying anti-cancer drug treatment: what can be seen from the proteomics point of view? *Cancer Treat. Rev.* 32, 619–629.
- Kim, Y.J., Shin, Y.K., Sohn, D.S., Lee, C.S., 2014. Menadione induces the formation of reactive oxygen species and depletion of GSH-mediated apoptosis and inhibits the FAK-mediated cell invasion. *Naunyn-Schmiedeberg Arch. Pharmacol.* 387, 799–809.
- Kolli-Bouhafs, K., Boukhari, A., Abusina, A., Velot, E., Gies, J.P., Lugnier, C., Rondé, P., 2012. Thymoquinone reduces migration and invasion of human glioblastoma cells associated with FAK, MMP-2 and MMP-9 down-regulation. *Invest. New Drugs* 30, 2121–2131.
- Li, Z.H., Bresnick, A.R., 2006. The S100A4 metastasis factor regulates cellular motility via a direct interaction with myosin-IIA. *Cancer Res.* 66, 5173–5180.
- Li, C.W., Xia, W., Huo, L., Lim, S.O., Wu, Y., Hsu, J.L., Chao, C.H., Yamaguchi, H., Yang, N.K., Ding, Q., Wang, Y., Lai, Y.J., LaBaff, A.M., Wu, T.J., Lin, B.R., Yang, M.H., Hortobagyi, G.N., Hung, M.C., 2012. Epithelial-mesenchymal transition induced by TNF- α requires NF- κ B-mediated transcriptional upregulation of Twist1. *Cancer Res.* 72, 1290–1300.
- Liew, K., Yong, P.V.C., Lim, Y.M., Navaratnam, V., Ho, A.S.H., 2014. 2-Methoxy-1,4-naphthoquinone (MNQ) suppresses the invasion and migration of a human metastatic breast cancer cell line (MDA-MB-231). *Toxicol. In Vitro* 28, 335–339.
- Lin, P.C., Chan, P.M., Hall, C., Manser, E., 2011. Collapsin response mediator proteins (CRMPs) are a new class of microtubule-associated protein (MAP) that selectively interacts with assembled microtubules via a taxol-sensitive binding interaction. *J. Biol. Chem.* 286, 41466–41478.
- Liu, L., Sun, L., Zhao, P., Yao, L., Jin, H., Liang, S., Wang, Y., Zhang, D., Pang, Y., Shi, Y., Chai, N., Zhang, H., Zhang, H., 2010a. Hypoxia promotes metastasis in human gastric cancer by up-regulating the 67-kDa laminin receptor. *Cancer Sci.* 101, 1653–1660.
- Liu, L., Zhang, H., Sun, L., Gao, Y., Jin, H., Liang, S., Wang, Y., Dong, M., Shi, Y., Li, Z., Fan, D., 2010b. ERK/MAPK activation involves hypoxia-induced MGR1-Ag/37LRP expression and contributes to apoptosis resistance in gastric cancer. *Int. J. Cancer* 127, 820–829.
- Lombaerts, M., van Wezel, T., Philippo, K., Dierssen, J.W.F., Zimmerman, R.M.E., Oosting, J., van Eijk, R., Eilers, P.H., van de Water, B., Cornelisse, C.J., Cleton-Jansen, A.M., 2006. E-cadherin transcriptional downregulation by promoter methylation but not mutation is related to epithelial-to-mesenchymal transition in breast cancer cell lines. *Br. J. Cancer* 94, 661–671.
- Lu, J.J., Bao, J.L., Wu, G.S., Xu, W.S., Huang, M.Q., Chen, X.P., Wang, Y.T., 2013. Quinones derived from plant secondary metabolites as anti-cancer agents. *Anticancer Agents Med. Chem.* 13, 456–463.
- Luanpitpong, S., Chanvorachote, P., Nimmannit, U., Leonard, S.S., Stehlik, C., Wang, L., Rojanasakul, Y., 2012. Mitochondrial superoxide mediates doxorubicin-induced keratinocyte apoptosis through oxidative modification of ERK and Bcl-2 ubiquitination. *Biochem. Pharmacol.* 83, 1643–1654.
- Martin, D., Salinas, M., Fujita, N., Tsuruo, T., Cuadrado, A., 2002. Ceramide and reactive oxygen species generated by H₂O₂ induce caspase-3-independent degradation of Akt/protein kinase B. *J. Biol. Chem.* 277, 42943–42952.
- Manu, K.A., Shanmugam, M.K., Rajendran, P., Li, F., Ramachandran, L., Hay, H.S., Kannaiyan, R., Swamy, S.N., Vali, S., Kapoor, S., Ramesh, B., Bist, P., Koay, E.S., Lim, L.H.K., Ahn, K.S., Kumar, A.P., Sethi, G., 2011. Plumbagin inhibits invasion and migration of breast and gastric cancer cells by downregulating the expression of chemokine receptor CXCR4. *Mol. Cancer* 10, 107.
- Matsuura, I., Lai, C.Y., Chiang, K.N., 2010. Functional interaction between Smad3 and S100A4 (metastatin-1) for TGF- β -mediated cancer cell invasiveness. *Biochem. J.* 426, 327–335.
- McDonald, K.K., Aulas, A., Destroismaisons, L., Pickles, S., Beleac, E., Camu, W., Rouleau, G.A., Vande Velde, C., 2011. TAR DNA-binding protein 43 (TDP-43) regulates stress granule dynamics via differential regulation of G3BP and TIA-1. *Hum. Mol. Genet.* 20, 1400–1410.
- Moriyama-Kita, M., Endo, Y., Yonemura, Y., Heizmann, C.W., Miyamori, H., Sato, H., Yamamoto, E., Sasaki, T., 2005. S100A4 regulates E-cadherin expression in oral squamous cell carcinoma. *Cancer Lett.* 230, 211–218.
- Nelson, J., McFerran, N.V., Pivato, G., Chambers, E., Doherty, C., Steele, D., Timson, D.J., 2008. The 67 kDa laminin receptor: structure, function and role in disease. *Biosci. Rep.* 28, 33–48.
- Pan, S., Wang, R., Zhou, X., He, G., Koomen, J., Kobayashi, R., Sun, L., Corvera, J., Gallick, G.E., Kuang, J., 2006. Involvement of the conserved adaptor protein Alix in actin cytoskeleton assembly. *J. Biol. Chem.* 281, 34640–34650.
- Pečina-Šlaus, N., 2003. Tumor suppressor gene E-cadherin and its role in normal and malignant cells. *Cancer Cell Int.* 3, 17.
- Ramos-DeSimone, N., Hahn-Dantona, E., Siple, J., Nagase, H., French, D.L., Quigley, J.P., 1999. Activation of matrix metalloproteinase-9 (MMP-9) via a converging plasmin/stromelysin-1 cascade enhances tumor cell invasion. *J. Biol. Chem.* 274, 13066–13076.

- Reynaert, N.L., van der Vliet, A., Guala, A.S., McGovern, T., Hristova, M., Pantano, C., Heintz, N.H., Heim, J., Ho, Y.S., Matthews, D.E., Wouters, E.F., Janssen-Heininger, Y.M., 2006. Dynamic redox control of NF-kappaB through glutaredoxin-regulated S-glutathionylation of inhibitory kappaB kinase beta. *Proc. Natl. Acad. Sci. U. S. A.* 103, 13086–13091.
- Seddighzadeh, M., Zhou, J., Kronenwett, U., Shoshan, M.C., Auer, G., Sten-Linder, M., Wiman, B., Linder, S., 1999. ERK signalling in metastatic human MDA-MB-231 breast carcinoma cells is adapted to obtain high urokinase expression and rapid cell proliferation. *Clin. Exp. Metastasis* 17, 649–654.
- Sethi, N., Kang, Y., 2011. Unravelling the complexity of metastasis – molecular understanding and targeted therapies. *Nat. Rev. Cancer* 11, 735–748.
- Shieh, J.M., Chiang, T.A., Chang, W.T., Chao, C.H., Lee, Y.C., Huang, G.Y., Shih, Y.X., Shih, Y.W., 2010. Plumbagin inhibits TPA-induced MMP-2 and u-PA expressions by reducing binding activities of NF- κ B and AP-1 via ERK signaling pathway in A549 human lung cancer cells. *Mol. Cell. Biochem.* 335, 181–193.
- Shih, Y.W., Lee, Y.C., Wu, P.F., Lee, Y.B., Chiang, T.A., 2009. Plumbagin inhibits invasion and migration of liver cancer HepG2 cells by decreasing productions of matrix metalloproteinase-2 and urokinase-plasminogen activator. *Hepatol. Res.* 39, 998–1009.
- Shu, L., Cheung, K., Khor, T.O., Chen, C., Kong, A., 2010. Phytochemicals: cancer chemoprevention and suppression of tumor onset and metastasis. *Cancer Metastasis Rev.* 29, 483–502.
- Tan, S.Y., 2011. Differential protein expression in human chronic myelogenous leukaemia cell line (K562) following exposure to 2-methoxy-1,4-naphthoquinone isolated from *Impatiens balsamina* Linn. MS Thesis dissertation. Department of Science, Faculty of Engineering and Science, Universiti Tunku Abdul Rahman, Kuala Lumpur, Malaysia.
- Tomsig, J.L., Sohma, H., Creutz, C.E., 2004. Calcium-dependent regulation of tumour necrosis factor-alpha receptor signalling by copine. *Biochem. J.* 378, 1089–1094.
- Tseng, H.C., Kao, H.W., Ho, M.R., Chen, Y.R., Lin, T.W., Lyu, P.C., Lin, W.C., 2011. Cytoskeleton network and cellular migration modulated by nuclear localized receptor tyrosine kinase ROR1. *Anticancer Res.* 31, 4239–4249.
- Venticinque, L., Jamieson, K.V., Meruelo, D., 2011. Interactions between laminin receptor and the cytoskeleton during translation and cell motility. *PLoS One* 6, e15895.
- Wang, Y.C., Lin, Y.H., 2012. Anti-gastric adenocarcinoma activity of 2-methoxy-1,4-naphthoquinone, an anti-*Helicobacter pylori* compound from *Impatiens balsamina* L. *Fitoterapia* 83, 1336–1344.
- Wang, H., Chunlian, W., Shengbang, W., Zhang, H., Zhou, S., Liu, G., 2013. Shikonin attenuates lung cancer cell adhesion to extracellular matrix and metastasis by inhibiting integrin β 1 expression and the ERK1/2 signaling pathway. *Toxicology* 308, 104–112.
- Wang, L., Zhang, D., Yu, Y., Guan, H., Qiao, C., Shang, T., 2013. RNA interference-mediated silencing of laminin receptor 1 (LR1) suppresses migration and invasion and down-regulates matrix metalloproteinase (MMP)-2 and MMP-9 in trophoblast cells: implication in the pathogenesis of preeclampsia. *J. Mol. Histol.* 44, 661–668.
- Weber, G.F., 2013. Why does cancer therapy lack effective anti-metastasis drugs? *Cancer Lett* 328, 207–211.
- Wu, Z.H., Chen, Z., Shen, Y., Huang, L.L., Jiang, P., 2011. Anti-metastasis effect of thymoquinone on human pancreatic cancer. *Yao Xue Xue Bao* 46, 910–914.
- Yan, J.X., Wait, R., Berkelman, T., Harry, R.A., Westbrook, J.A., Wheeler, C.H., Dunn, M.J., 2000. A modified silver staining protocol for visualization of proteins compatible with matrix-assisted laser desorption/ionization and electrospray ionization-mass spectrometry. *Electrophoresis* 22, 3666–3672.
- Yan, C., Boyd, D.D., 2007. Regulation of matrix metalloproteinase gene expression. *J. Cell. Physiol.* 211, 19–26.
- Yang, Q., Ji, M., Guan, H., Shi, B., Hou, P., 2013. Shikonin inhibits thyroid cancer cell growth and invasiveness through targeting major signaling pathways. *J. Clin. Endocrinol. Metab.* 98, E1909–E1917.
- Yilmaz, M., Christofori, G., 2009. EMT, the cytoskeleton, and cancer cell invasion. *Cancer Metastasis Rev.* 28, 15–33.
- Yonemura, Y., Endou, Y., Kimura, K., Fushida, S., Bandou, E., Taniguchi, K., Kinoshita, K., Ninomiya, I., Sugiyama, K., Heizmann, C.W., Schafer, B.W., Sasaki, T., 2000. Inverse expression of S100A4 and E-cadherin is associated with metastatic potential in gastric cancer. *Clin. Cancer Res.* 6, 4234–4242.
- Zhang, G., Kernan, K.A., Collins, S.J., Cai, X., López-Guisa, J.M., Degen, J.L., Shvil, Y., Eddy, A.A., 2007. Plasmin(ogen) promotes renal interstitial fibrosis by promoting epithelial-to-mesenchymal transition: role of plasmin-activated signals. *J. Am. Soc. Nephrol.* 18, 846–859.
- Zhang, J., Zhang, D.L., Jiao, X.L., Dong, Q., 2013. S100A4 regulates migration and invasion in hepatocellular carcinoma HepG2 cells via NF- κ B-dependent MMP-9 signal. *Eur. Rev. Med. Pharmacol. Sci.* 17, 2372–2382.